Likelihood

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- If you are familiar Bayesian statistics...

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- For <reasons> people also write likelihood as $\mathcal{L}\left(\vec{\theta} \mid \vec{x}\right) \equiv \Pr\left(\vec{x} \mid \vec{\theta}\right)$
  - In this case, the bar on the LHS no longer denotes "condition"

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- From the frequentist viewpoint, we assume there is a (fixed) underlying truth θ_{truth} which "best explains" the measured data $\vec{x}$

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- If there is no signal in our data ($\theta_{\text{truth}} = \theta_0$), then the ratio is 1.

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- Because <nice math property>, we tag a 2 to the log ratio

$$\Lambda = 2 \times \ln\left[\frac{\mathcal{L}(\theta)}{\mathcal{L}(\theta_0)}\right]$$

where Λ is known as the **test statistic**

Wilk's Theorem and p-value

- We would then try to maximize the test statistic

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